

Brijodison

Engineering Notebook Element C

Engineering Service Learning

Criteria Ranked by Importance

- 1. Ability to Push (most important)** - Our product won't be useful at all if it takes away from the original function of the cart. If the cart is no longer functional as a result of the handle, the user's back will see no improvement. The original handle on the cart is very usable when pushing the cart, so making an item that doesn't keep or add usability to the cart handle would not be beneficial. In particular, we can measure the ability to push the cart by measuring the force (with a force plate) required to budge the cart. First and foremost, the users of the product want a cart that functions, so that is our first priority. Mr. Kiser and the robotics students (both primary stakeholders) use this cart to push goods, so it is most important that we maintain that use case.
- 2. Bending to Build** - We think that this is another important feature required to make this a successful product. When making this design we want to make a product that can be easily attached and detached from the cart without ignoring the intention of the product itself. If the user has to bend down in an uncomfortable manner to attach our cart handle extender, the bending defeats the main purpose of the product—to prevent back pain by limiting bending. We intend to measure this criteria by measuring the maximum angle at which someone must bend over to assemble the product. Mr. Kiser agreed with the notion

that assembly should be easy, and further discussion with him as a primary stakeholder was used to build the range for this criteria in our decision matrix.

- 3. Stability** - After speaking with Jared Lawson, a graduate student at Vanderbilt with a B.S. in Mechanical Engineering, we chose to add a form of stability as a criteria. This criteria fits well with the “Ability to Push” criteria, but the key difference is it applies more to turning. A proper solution should turn the cart without significantly moving the handle. To objectively measure this, we can use a force plate at a specified distance to determine the force needed to budge the handle.
- 4. Height Range** - One of the main problems with the original handle is that it is not high enough. We want to make a design that can either be used at multiple heights or to be adjusted to a height that is comfortable for the user. We intend to measure the final height range of the product by measuring the height difference between the highest position the handle may be adjusted to and the lowest position. Mr. Kiser, a primary stakeholder, noted that any product would have to be at least minutely adjustable to work for him. We are also aware that other primary stakeholders (e.g. robotics students) will require height adjustment to use the product. We organize this priority lower to account for its intention: comfort. The first priorities determine the usability of the product, but this criteria is the most important for comfortability because it dictates whether minimal strain is placed.

5. Time to Assemble and Disassemble - Mr. Kiser, a primary stakeholder, mentioned the importance of ease-of-assembly. We have already introduced criteria to ensure the assembly will not place more strain on the stakeholder, but we would now like to assess the general ability to assemble and disassemble the product. We can measure the time it takes multiple (e.g. five) people to assemble the product and subsequently disassemble it without rest to objectively find this information. By taking the average, we will obtain a number that may be used as a proxy for the ease of assembly and disassembly. We chose to combine these criteria into one because they act together. If the product is difficult to assemble but easy to disassemble, the stakeholder is unlikely to take it apart with the knowledge they will eventually have to reassemble it. On the other hand, if the product is easy to assemble but difficult to disassemble, the stakeholder is unlikely to disassemble it for sheer difficulty, so the assembly time has no significance.

6. Weight - This is a very important part that we have to take into account while designing. How heavy our product weighs could either benefit or damage our idea. With a heavier weight it would hold on to the cart better but it might be harder to attach and move different places. If the product is too heavy, it will not be useful because it will place too much strain on the primary stakeholder's back, counteracting the benefit from its usage. We will be able to easily and objectively measure the weight with a scale.

7. Size - If the product is difficult to carry around, the stakeholder is less likely to use it. We are already aware from discussions with Mr. Kiser that he will want to remove the

product, which implies he will have to store the product somewhere. This storage will be far more difficult if the product is oversized. Its size may also cause vision impairment that will hurt the cart's ability to function. We rank this lower than the above criteria because it will likely have little impact on the stakeholder's back, thus satisfying something beyond the very core of the problem statement. We will measure size objectively as the smallest box within which the product could fit (note this is different from its volume because we must consider the overall space it takes up when stored in a box for instance is more important in context).

- 8. Ergonomic/comfortability** - Though this is not at the top of the list we think that it is still a very important criteria to consider when making the cart handle. We want to make a design that works but that is also comfortable while using. While making our product our plan would be to use it multiple times a day with ease and comfortability. The competition is not particularly difficult as the current solution is a rigid plastic handle, so our minimum should be to not make the handle less comfortable. As mentioned by Mr. Kiser, the handle does have built in ergonomic features, so we would like to continue these in some way. We have placed this criteria towards the bottom of the list because it is very difficult to objectively measure, but we can allow multiple people to use the cart and rank the handle's comfort. To get more objective scores, we will ask them to use the standard handle first, calling it a five out of ten as a baseline for comfort. We will then ask them to rank the novel handle in comparison (out of ten). By asking them to compare, we reduce the opinionated portion of this criteria. We also standardize their responses by

giving them a baseline. We will take the average of multiple stakeholders to aid objectivity.

9. Expenses - While a constraint, cost is also a criteria. There is a hard budget of \$30, but we would also ideally reduce the money spent on the product. If we spend less money on its production, customers are more likely to purchase it for a subsequently lower price. The cost, however, is subservient to the quality of the product to a point (and that point is the constraint), which is why it is ranked last. We will measure this objectively as the price of materials needed to create the product and an estimate of labor costs found by taking the total time used to create the product and multiplying by 25.76 dollars/hour (the average hourly rate for a machinist) (Indeed).

10. Aesthetics (least important) - Our lowest criteria is the aesthetics of the product. It is very difficult to objectively measure the aesthetics, and it does not impact the functionality of the product at all, so we are less concerned about it as a criteria. Our primary stakeholder, however, has mentioned that it has some significance to him, so we should do our due diligence to include it. To attempt to objectively measure the aesthetics of the product in comparison to the cart (because the two must be paired together, we should ensure they fit aesthetically), we may ask a variety of people to look at the cart alone and consider it a five out of ten. We will then add the product, and ask the audience to rank the combination out of ten. We will average the results to add objectivity. The comparison is also in hopes of adding objectivity.

Works Cited

“Machinist Salary in United States.” *Indeed.com*, 2024,
www.indeed.com/career/machinist/salaries.